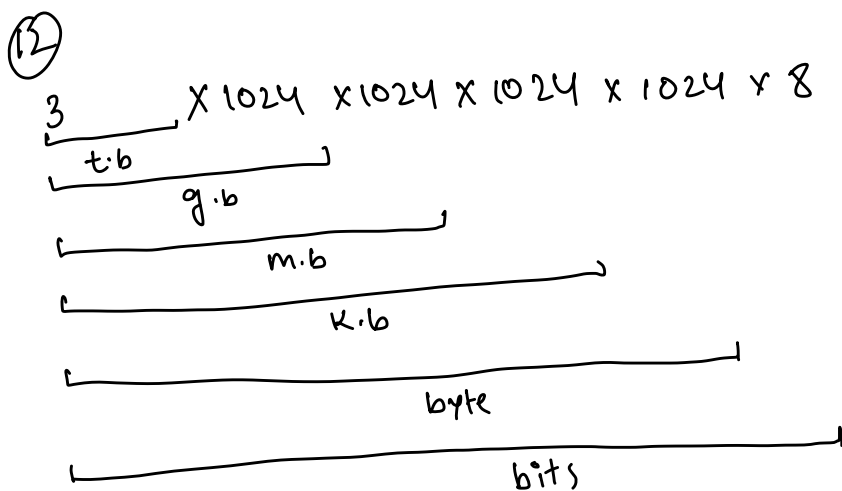


- 1 How many address lines do we need for a 64 MB RAM with 16 bit/words?
- 2 Design a 2048x32 RAM showing internal details. *Slide 3 and 4*
- 3 Design a 4096x16 RAM showing internal details.
- 4 What is the capacity for a $2^{16} \times 16$ RAM in Gigabytes?
- 5 Draw the block diagram of a binary cell.
- 6 Draw and explain the functionalities of MAR and MBR.
- 7 What type of information can you get from the "2048x32" part of a 2048x32 RAM? *Ans: 8 kb memory storing 32 bit values.*
- 8 How many types of RAM are there? *Ans: 2*
- 9 What is static RAM? : *use flip flop to represent data*
- 10 What is dynamic RAM? : *use capacitor charges to represent data*
- 11 Convert 8,796,093,022,208 Bits to Gigabytes.
- 12 Convert 3 Terabytes to bits.

$$\begin{aligned} \textcircled{1} \quad 64 \text{ MB} \times 1024 &= 65536 \text{ kilo byte} \\ &= 65536 \times 8 \times 1024 \text{ bits} \\ &= 536870912 \text{ bits} \end{aligned}$$

$$\begin{aligned} \therefore 16 \text{ bits} &= 1 \text{ word} \\ \therefore 536870912 \text{ bits} &= \frac{536870912}{16} \text{ word} \\ &= 33554432 \text{ words.} \end{aligned}$$

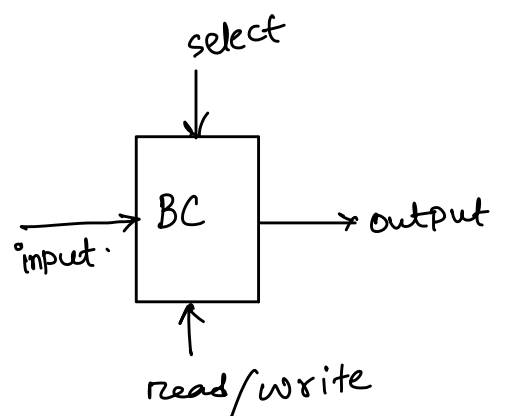
$$\begin{aligned} \therefore 33554432 \text{ word} &= \log_2(33554432) \text{ address lines} \\ &= 25 \text{ address lines.} \end{aligned}$$



$$= 2.6388279 \times 10^{13} \text{ bits}$$

$$\begin{aligned} \textcircled{11} \quad &\frac{8796093022208}{8} \text{ byte} \\ &= 1099511627766 \text{ byte} \\ &= \frac{1099511627766}{1024} \text{ k.byte} \\ &= 1073741824 \text{ K.byte} \\ &= \frac{1073741824}{1024} \text{ Mega byte} \\ &= 1048576 \text{ megabyte} \\ &= \frac{1048576}{1024} \text{ Gigabyte} \\ &= 1024 \text{ gigabyte} \end{aligned}$$

$\textcircled{5}$ Diagram of a binary cell



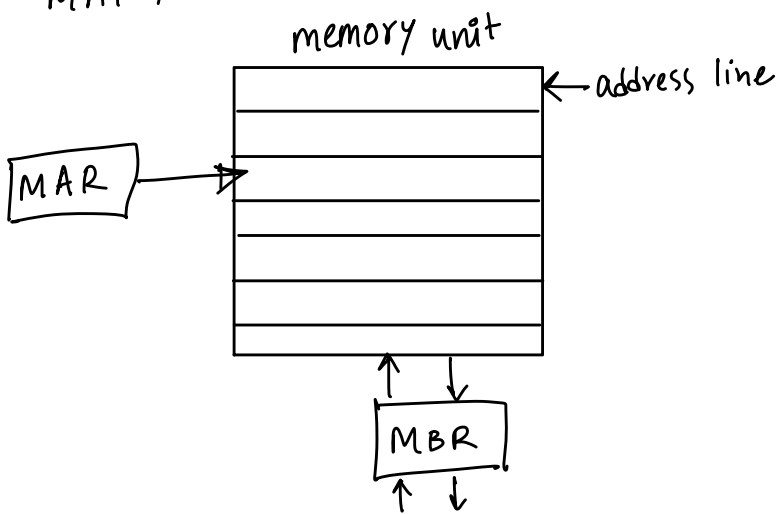
④ no of words = 2^{16}
bits per word = 16

$\therefore$ 1 word = 16 bits

$\therefore 2^{16}$ words = $2^{16} \times 16$ bits

$$= 1048576 \text{ bits} = \frac{1048576}{8} \text{ byte} = \frac{131072}{1024} \text{ kilobyte}$$

⑥ MAR, MBR



$$= \frac{128}{1024} \text{ megabyte}$$

$$= \frac{0.125}{1024} \text{ gigabyte}$$

$$= 0.00012207031 \text{ gigabyte}$$

Memory Address Register [MAR]: addressed transferred $\rightarrow$ MAR $\rightarrow$ find and open path to line

Memory Buffer Register [MBR]: before writing and after reading infos are kept. To ensure memory does not remain idle.

③ 4096x16

